**13.2.2 Exercises - Gravitational Field 2**

1.

a. Define gravitational field strength.

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b. A spherical planet has diameter $1.2 \times 10^4$ km. The gravitational field strength at the surface of the planet is 8.6 N $kg^{-1}$. The planet may be assumed to be isolated in space and to have its mass concentrated at its centre. Calculate the mass of the planet.

mass = ..... kg [3]

c. The gravitational potential at a point X above the surface of the planet in (b) is $-5.3 \times 10^7 J\ kg^{-1}$. For point Y above the surface of the planet, the gravitational potential is $-6.8 \times 10^7 J\ kg^{-1}$.

i. State, with a reason, whether point X or point Y is nearer to the planet.

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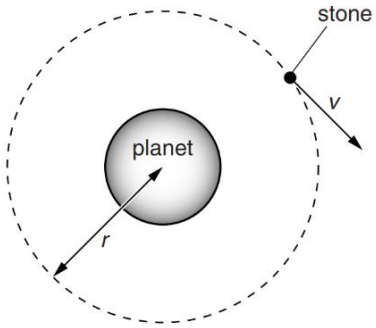
.....[2]

ii. A rock falls radially from rest towards the planet from one point to the other. Calculate the final speed of the rock.

speed = ..... m $s^{-1}$ [2]

2. The mass M of a spherical planet may be assumed to be a point mass at the centre of the planet.

a. A stone, travelling at speed v, is in a circular orbit of radius r about the planet,



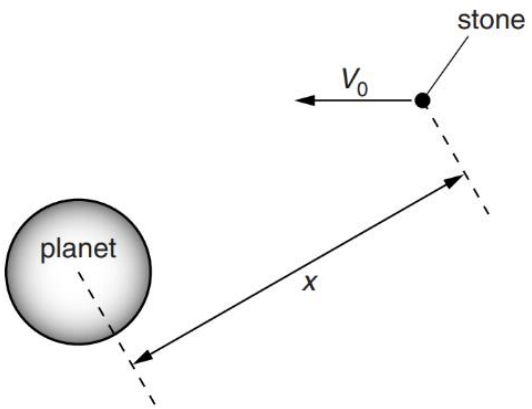
Show that the speed v is given by the expression

$$v = \sqrt{\frac{GM}{r}}$$

where G is the gravitational constant.

Explain your working. [2]

b. A second stone, initially at rest at infinity, travels towards the planet,




The stone does not hit the surface of the planet.

i. Determine, in terms of the gravitational constant G and the mass M of the planet, the speed V0 of the stone at a distance x from the centre of the planet. Explain your working. You may assume that the gravitational attraction on the stone is due only to the planet. [3]

ii. Use your answer in (i) and the expression in (a) to explain whether this stone could enter a circular orbit about the planet.

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.....[2]

3.

a. Define gravitational potential at a point.

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.....[2]

b. The Moon may be considered to be an isolated sphere of radius $1.74 \times 10^3$ km with its mass of $7.35 \times 10^{22}$ kg concentrated at its centre.

i. A rock of mass 4.50 kg is situated on the surface of the Moon. Show that the change in gravitational potential energy of the rock in moving it from the Moon's surface to infinity is $1.27 \times 10^7$ J. [1]

ii. The escape speed of the rock is the minimum speed that the rock must be given when it is on the Moon's surface so that it can escape to infinity.

Use the answer in (i) to determine the escape speed. Explain your working.

speed = ..... m $s^{-1}$ [2]

c. The Moon in (b) is assumed to be isolated in space. The Moon does, in fact, orbit the Earth. State and explain whether the minimum speed for the rock to reach the Earth from the surface of the Moon is different from the escape speed calculated in (b).

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.....[2]

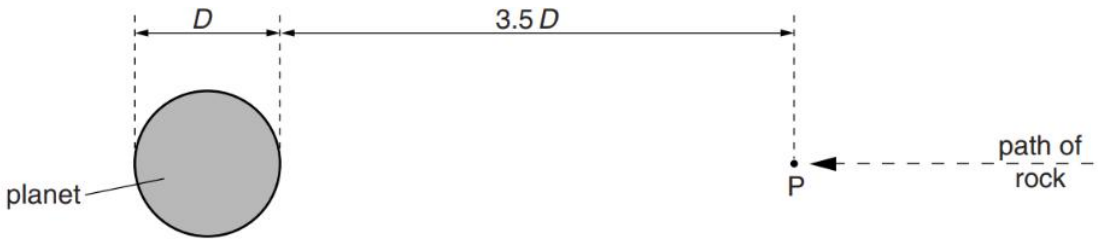
4. An isolated spherical planet has a diameter of $6.8 \times 10^6$ m. Its mass of $6.4 \times 10^{23}$ kg may be assumed to be a point mass at the centre of the planet.

a. Show that the gravitational field strength at the surface of the planet is 3.7 N $kg^{-1}$. [2]

b. A stone of mass 2.4 kg is raised from the surface of the planet through a vertical height of 1800 m. Use the value of field strength given in (a) to determine the change in gravitational potential energy of the stone. Explain your working.

change in energy = ..... J [3]

c. A rock, initially at rest at infinity, moves towards the planet. At point P, its height above the surface of the planet is 3.5 D, where D is the diameter of the planet,




Calculate the speed of the rock at point P, assuming that the change in gravitational potential energy is all transferred to kinetic energy.

speed = ..... m $s^{-1}$ [4]